

A novel improved Frilled Lizard algorithm for solving the optimal planning problem of renewable energy sources within distribution grids under uncertainties

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ABSTRACT

In this research, a new improved frilled lizard optimization algorithm is designed and proposed to address the optimal planning problem of renewable energy sources-based distributed generation within radial distribution grids. The primary objective is to minimize power losses, elevate the voltage profile, and improve voltage stability, formulated as a multi-objective optimization problem. The planning process of renewable energy sources in the radial distribution grid is a complex problem that requires accounting for uncertainties in renewable energy sources' power output and load demand fluctuations. To this end, we have developed an appropriate probability model to estimate the stochastic power generation from renewable energy sources based on hourly seasonal data, including wind speed, solar irradiance, and ambient temperature collected over a specified time frame and location. The developed improved frilled lizard optimization technique incorporates three distinct strategies: fitness distance balance, quasi-opposite-based learning, and Cauchy mutation, to enhance its searching capabilities and avoid falling into local optimal traps. The suggested improved frilled lizard optimization is successfully applied to identify the optimal locations and rating capacities of solar photovoltaic strings and wind turbines, as well as the power factor of wind turbines. The effectiveness of the presented approach is demonstrated using the IEEE 85-bus distribution grid. To further assess its feasibility and robustness, a performance comparison is conducted against other recent effective algorithms, including the grey wolf optimizer, the black-winged kite algorithm, and the original frilled lizard optimization. Results show that the proposed improved frilled lizard optimization technique significantly reduces annual average power losses by 70.25%, decreases voltage deviation by 78.51%, and improves the voltage stability index by 20.79%, outperforming other methods.

1. Introduction

Recently, the surge in electricity demand has become increasingly evident due to economic development and the installation of new facilities to accommodate the growing population. The traditional response to meet this higher demand involves augmenting the power plant capacities through further combustion. However, this approach is neither economically effective nor environmentally sound, as it relies heavily on fossil fuels [1]. Moreover, the rising loads introduce potential operational issues for the distribution grid, such as increased power losses, voltage drops, degradation in power quality, and deterioration in system stability, particularly due to its radial structure. [2,3].

Renewable energy sources (RESs) such as solar photovoltaics (PVs) systems and wind turbines (WTs) based distributed generations (DGs)

offer an alternative solution to meet the growing demand for electricity and enhance the performance of radial distribution grids (RDGs). RESs are cost-effective and environmentally friendly, referring to small-scale electricity generation, placed near the end-consumers [4,5]. Integrating RESs into the RDG offers substantial benefits, including, reduced power losses, improved voltage profile, enhanced system stability, decreased carbon gas emissions and alleviated heavy loads. However, these benefits cannot be earned until DGs are strategically located and optimally sized [6]. Improper placement and sizing of DGs can negatively impact system performance, such as increased power losses, poor voltage profiles, system instability, reverse power flow and improper functioning of protection devices [7,8]. Therefore, effectively integrating RESs into RDGs presents a crucial challenge for many researchers and engineers.

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Nonetheless, the intermittent nature and unpredictability of RESs introduce additional challenges and intricacies in optimal grid planning. Wind and solar power generation are particularly affected by meteorological conditions. WTs depend on wind speed, while PV systems are influenced by solar irradiance and ambient temperature at a specific time and a geographic location. These dependencies, coupled with fluctuations in load demand, complicate the proper incorporation of RESs into the RDG. Addressing these challenges is a crucial step toward maximizing the effectiveness of RES integration and ensuring a reliable and sustainable power system [9].

In the academic literature, many studies have attempted to solve the optimal planning problem (OPP) of DGs within the RDG, each considering specific objectives, either as a single or multi-objective problem formulation. These studies involve the application of different optimization techniques, encompassing the whale optimization algorithm (WOA) [10], the equilibrium optimizer (EO) algorithm [11], the improved wild horse optimization (IWHO) algorithm [12] and the adaptive particle swarm optimization (APSO) [13] to identify the optimal size and site of DGs for reducing power loss. Also, the coyote optimization algorithm (COA) [14] has been applied to find the best site and size of DG units to reduce power loss, lower operating costs, and improve voltage stability. The modified gravitational search algorithm (MGSA) [15], the artificial bee colony (ABC) algorithm [3] and the arithmetic optimization algorithm (AOA) [16] have been employed to identify the best site and location of DG units for reducing power losses and elevate the voltage profile. Furthermore, the quasi oppositional chaotic symbiotic organisms search (QOCSOS) algorithm [5] and the grey wolf optimization (GWO) [17] were applied to pinpoint the optimal size and site of DG units to reduce power loss, improve voltage profile, and enhance system stability.

Overall, the above studies focused on addressing the DG sizing and location problem by considering peak load conditions. Additionally, the uncertainties associated with renewable sources have been omitted. However, other studies have addressed the integration of renewable energy sources, considering their intermittent nature and variable load demands within the distribution grid from various perspectives. For example Kayal and Chanda [18] focused on optimizing the locations and capacities of mixed solar and wind energy sources in an Indian rural 28-bus distribution grid. The weighted aggregation particle swarm optimization (WAPSO) algorithm was utilized to reduce power losses, enhance voltage stability, and improve network security while accounting for uncertainties in solar irradiance and wind speed. Hemeida et al. [19], presented a methodology for optimizing DGs placement in power systems considering load uncertainties. The approach utilizes Monte Carlo simulation to generate various load scenarios employing four bio-inspired optimization techniques, namely grey wolf optimization (GWO), manta ray foraging optimization (MRFO), satin bowerbird optimization (SBO), and whale optimization algorithm (WOA). The objective was to reduce active power loss, voltage deviation, and maximize the voltage stability index. In [20], the authors suggested a strategy for allocating PV and battery energy storage systems using a modified Henry gas solubility optimization (HGSO) algorithm to minimize total system power loss, considering PV generation uncertainties and time-varying loads. In [21], the authors introduced a novel approach employing genetic algorithms (GA) to optimize the allocation of capacitor banks (CB) and distributed PV generation in unbalanced distribution grids. The method included conducting sensitivity analysis using GA to pinpoint optimal allocation zones where placing PV and CB could minimize losses. The study customized GA to manage uncertainties in load and PV generation over a 24-hour planning horizon. Khasanov et al. [22], suggested an artificial ecosystem-based optimization–opposition-based learning (AEO-OBL), to optimize the placement of PV and WT-based DGs on IEEE 33-bus and 85-bus RDGs. The improved algorithm incorporates five strategies to enhance the search capability and achieve a balance between the exploration and

exploitation phases, addressing the stochastic nature of RESs and making it effective for minimizing system power losses. In [23], the rider optimization algorithm (ROA) has been proposed to identify the optimal size and site of RES, specifically WTs, PVs and biomass-based DG units in standard 33 and 69-bus distribution grids. The study aimed to minimize total energy losses while considering the fluctuations in wind speed and solar irradiation. The authors of [24] introduces the artificial hummingbird algorithm (AHA) for optimal sizing and positions of RESs in IEEE 33-bus and actual 94-bus RDGs, effectively minimizing costs, emissions, and voltage deviations while enhancing voltage stability, considering uncertainties. A pseudo-inspired gravitational search algorithm (PI-GSA) is introduced in [25] as a novel approach to optimize the allocation and sizing of RERs in IEEE 15-bus, 33-bus and 69-bus distribution grids. The algorithm considered uncertainties related to PV and wind output, electricity prices, and load variations using probability distribution functions. In [26], a hybrid model for integrating renewable energy sources into the IEEE 33-bus distribution grid is proposed, using a bi-directional long short-term memory network to manage uncertainties from PV and wind generation. The model, optimized with binary particle swarm optimization (BPSO), aimed to minimize energy losses and voltage deviations while considering long-term and seasonal impacts. Most of these studies have addressed the OPP of RESs while considering load, solar irradiation, and wind speed uncertainties. However, they have largely overlooked temperature-related uncertainties, which limits their scope. Additional limitations in the existing literature include the assumption of fixed locations or sizes for RESs, as well as the neglect of certain grid constraints. Moreover, the majority of these approaches assumed that WTs operate at unity power factors (PFs). Consequently, these studies focus exclusively on the location and size of the WTs, neglecting the optimization of their PF, which is crucial for enhancing system performance and overall grid efficiency. Table 1 summarizes the current work and other relevant research on the OPP of RESs within RDG.

In this paper, we present a novel optimization approach, termed the Improved Frilled Lizard (IFLO) algorithm, designed to simultaneously determine the optimal positions and number of WTs and PV strings, as well as the optimal PF of WTs, within the 85-bus radial distribution grid. Our primary objective includes minimizing average annual power losses and voltage deviation (VD) while maximizing the yearly average voltage stability index (VSI) within operational constraints. The study considers uncertainties in RES, including fluctuations in wind speed, availability of solar irradiance, temperature variations, and time-varying load demand, based on real historical collected data from the study area over a seasonal planning horizon. The proposed approach is evaluated against several established optimization techniques, including the well-known Grey Wolf Optimizer (GWO) [27], Harris Hawks optimization (HHO) [28], Chernobyl Disaster Optimizer (CDO) [29], Dingo Optimization Algorithm (DOA) [30], Whale Optimization Algorithm (WOA) [31], Black-winged Kite Algorithm (BKA) [32] and the original FLO [33]. However, the novelty and main achievements of this research are summarized below:

1. A novel Improved Frilled Lizard (IFLO) Algorithm is developed and proposed to solve the optimal planning problem (OPP) of RESs within the radial distribution grids (RDGs).
2. A more realistic approach based on Beta and Weibull probability density functions (PDFs) is operated to model the stochastic behavior of PV and WT power generation, respectively, based on real historical data collected over a decade from the study area.
3. The optimal locations and number of WTs and PV strings, as well as the PF of WTs, are determined simultaneously under seasonal uncertainties. This aims to reduce average annual power losses and voltage deviation (VD) while enhancing the yearly average voltage stability index (VSI), through a multi-objective optimization problem formulation.

Table 1

Comparison analysis between the current study and other relevant works for the OPP of RESs within RDG.

Ref.	Year	Solving method	Improved version	Technical objectives			Installation		Uncertainty			
				Power losses	Voltage deviation	Voltage stability	WT	PV	Loading	Wind speed	Solar irradiance	Temperature
[13]	2023	APSO	✓	✓	×	×	✓	×	×	×	×	×
[18]	2015	WAPSO	×	✓	×	✓	✓	✓	✓	✓	✓	×
[19]	2021	MRFO	×	✓	✓	✓	✓	×	✓	×	×	×
[20]	2021	modified HGSO	✓	✓	×	×	×	✓	✓	×	✓	×
[21]	2022	GA	×	✓	×	×	×	✓	✓	×	✓	×
[23]	2021	ROA	×	✓	×	×	✓	✓	✓	✓	✓	×
[24]	2023	AHA	×	×	✓	✓	✓	✓	✓	✓	✓	×
[22]	2023	AEO-OBL	✓	✓	×	×	✓	✓	✓	✓	✓	×
[17]	2023	GWO	×	✓	✓	×	✓	×	×	×	×	×
[25]	2024	PI-GSA	✓	✓	✓	×	✓	✓	✓	✓	✓	×
[26]	2024	BPSO	✓	✓	✓	×	✓	✓	✓	✓	✓	×
This study		IFLO	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

Notation: Each column corresponds to a specific feature. The symbol “✓” indicates that the paper includes the feature, while the “×” signifies its absence.

- The effectiveness and reliability of the IFLO algorithm are thoroughly assessed via extensive statistical comparison with other efficient optimization techniques using standard benchmark functions.
- The proposed IFLO outperforms other developed optimization techniques, namely GWO, BKA, and the original FLO, in solving the optimal planning problem of RESs in the IEEE 85-bus RDG, demonstrating its effectiveness in dealing with uncertainties within a complex optimization framework.
- To the best of the authors' knowledge, no one has considered the uncertainties in the ambient temperature which has a major influence on PV systems.

The subsequent parts of the paper are organized as follows: Section 2 introduces the probabilistic system parameters model, which includes both load and RESs. Section 3 provides a detailed mathematical formulation of the objective functions and constraints. The design of the FLO and the proposed IFLO algorithms is described in Sections 4 and 5, respectively. The analysis of load profiles and weather data is discussed in Section 6. Section 7 presents the simulation results and discussion. Finally, Section 8 concludes the paper and outlines future research directions.

2. System modeling

This section details the modeling approaches for each uncertain parameter. The study considers time-varying constant load demand. Since power generated by WTs and solar PV systems depends on meteorological factors like wind speed, solar irradiance, and ambient temperature, the stochastic behavior of these factors must be thoroughly analyzed during the initial phase to ensure efficient integration of WT and PV systems.

2.1. Load modeling

The system being studied is presumed to adhere to the time-varying load curve of the IEEE-RTS [34,35]. To analyze seasonal variations in demand, the yearly load is segmented into four seasons: summer, autumn, winter, and spring. Each load season is represented by the hourly load of each day (24 h). Thus, with four season days, this results in 96 h (24 × 4), representing the entire year load (8760 h).

2.2. Renewable energy resources (RESs) modeling

2.2.1. Wind speed modeling

The stochastic behavior of wind speed over a specified time period is described using the Weibull PDF [36,37]. The following equation

expresses the Weibull PDF of wind speed w^t in (m/s) at t th hour.

$$f_w^t(w) = \left(\frac{k^t}{c^t}\right) \left(\frac{w^t}{c^t}\right)^{(k^t-1)} \times \exp\left[-\left(\frac{w^t}{c^t}\right)^{k^t}\right] \text{ for } w^t > 0, c^t > 0 \text{ and } k^t > 0 \quad (1)$$

The shape factor (k^t) and scale factor (c^t) at t th hour are computed as below.

$$k^t = \left(\frac{\sigma_w^t}{\mu_w^t}\right)^{-1.086} \quad (2)$$

$$c^t = \frac{\mu_w^t}{\Gamma(1 + 1/k^t)} \quad (3)$$

here, μ_w^t and σ_w^t symbolize the mean and standard deviation of wind speed at the t th hour.

2.2.2. Solar irradiance modeling

The random nature of solar irradiance is represented using the beta PDF [36,38]. The beta PDF of solar irradiance i^t in (kW/m²) over a timeframe ' t ' is expressed as follows,

$$f_i^t(i) = \begin{cases} \frac{\Gamma(\alpha^t + \beta^t)}{\Gamma(\alpha^t)\Gamma(\beta^t)} (i^t)^{\alpha^t-1} (1 - i^t)^{\beta^t-1}, & 0 \leq i^t \leq 1; \alpha^t, \beta^t \geq 0 \\ 0, & \text{otherwise} \end{cases} \quad (4)$$

where α^t and β^t denote the shape factors of beta PDF and Γ represents the gamma function.

The shape factors of beta PDF can be computed using the mean (μ_i^t) and the standard deviation (σ_i^t) of solar irradiance for the pertinent period.

$$\beta^t = (1 - \mu_i^t) \left(\frac{\mu_i^t(1 + \mu_i^t)}{(\sigma_i^t)^2} - 1 \right) \quad (5)$$

$$\alpha^t = \mu_i^t \times \frac{\beta^t}{1 - \mu_i^t} \quad (6)$$

2.2.3. Power generation model

To accurately estimate the power produced from the WT and solar PV, the continuous PDF for a specified timeframe is segmented into intervals (spans), where wind speed and solar irradiance are within specific ranges [18]. Thus, the power outputs from WT and solar PV are determined according to all possible probabilities within that hour.

2.2.3.1. Power generation by WT.

The average power output of the WT that corresponds to a particular timeframe ' t ', (P_{WT}^t) is calculated as follows:

$$P_{WT}^t = \sum_{k=1}^{N_w} PG_{WTk} \times P_w(w_k^t) \quad (7)$$

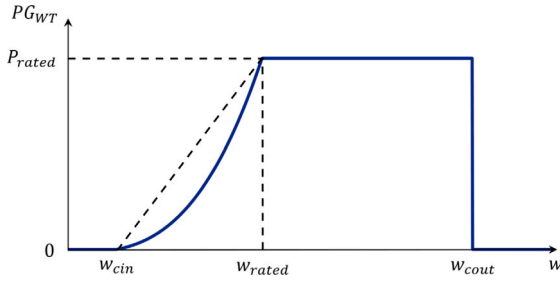


Fig. 1. Non-linear characteristics curve of wind power.

where 'k' signifies the phase variable and N_w denotes the number of discrete wind speed phases. w_k^t is the k th phase of wind speed at t th hour.

The probability of wind speed for each phase 'k' during any particular hour is estimated using,

$$P_w(w_k^t) = \int_{w_{k,min}^t}^{w_{k,max}^t} f_w^t(w) dw \quad \text{for } k = 1 \dots N_w \quad (8)$$

The wind speed at a WT's location determines its power output, which adheres to the non-linear performance characteristics set by the manufacturer. The generated power by the WT at average wind speed (w_{ak}) for the k th phase can be computed using [36]:

$$PG_{WTk} = \begin{cases} 0 & w_{ak} < w_{cin} \text{ or } w_{ak} > w_{cout} \\ (\alpha \times w_{ak}^3 + \beta \times P_{rated}) & w_{cin} \leq w_{ak} \leq w_N \\ P_{rated} & w_N \leq w_{ak} \leq w_{cout} \end{cases} \quad (9)$$

where P_{rated} is the maximum power output of the WT, w_{cout} denotes the cut-out wind speed. The constants α and β are determined based on the cut-in wind speed w_{cin} and the nominal wind speed w_n as follows,

$$\alpha = \frac{P_{rated}}{(w_n^3 - w_{cin}^3)} \quad (10)$$

$$\beta = \frac{w_{cin}^3}{(w_{cin}^3 - w_n^3)} \quad (11)$$

The non-linear performance characteristics curve of the WT is displayed in Fig. 1.

2.2.3.2. Power generation by solar PV.

The average power output of solar PV that corresponds to a particular timeframe 't', (P_{PV}^t) can be computed using,

$$P_{PV}^t = \sum_{k=1}^{N_i} PG_{PVk} \times P_i(i_k^t) \quad (12)$$

where N_i denotes the number of discrete solar irradiance phases and i_k^t is the k th phase of solar irradiance at t th hour.

The probability of solar irradiance for each phase 'k' during any particular hour is estimated using,

$$P_i(i_k^t) = \int_{i_{k,min}^t}^{i_{k,max}^t} f_i^t(i) di \quad (13)$$

The power output of solar PV arrays is influenced by the level of solar irradiance and the ambient temperature at the site where the modules are installed. The generated power by solar PV panels at average solar irradiance (i_{ak}) for the k th phase can be evaluated as [34] in Eq. (14).

$$PG_{PVk}(i_{ak}) = N_{PVmod} \times FF \times V_k \times I_k \quad (14)$$

where N_{PVmod} is the total PV modules number; FF is the fill factor; V_k , I_k are the cell voltage and the cell current at the k th phase, respectively.

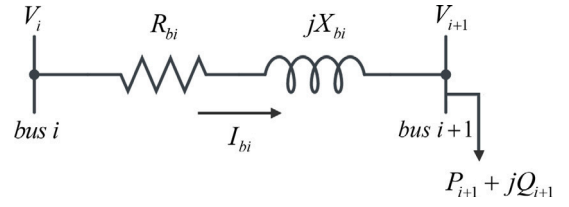


Fig. 2. Equivalent two-bus model of the grid.

These parameters can be calculated using the following formulas:

$$FF = \frac{V_{MMP} \times I_{MMP}}{V_{oc} \times I_{sc}} \quad (15)$$

$$V_k = V_{oc} - K_v \times T_{ck} \quad (16)$$

$$I_k = i_{ak}[I_{sc} + K_i(T_{ck} - 25)] \quad (17)$$

$$T_{ck} = T_a^t + i_{ak} \left(\frac{N_{OT} - 20}{0.8} \right) \quad (18)$$

here, V_{MMP} and I_{MMP} are respectively the voltage (V) and current (A) at maximum power point; V_{oc} and I_{sc} are respectively the open circuit voltage (V) and the short circuit current (A); T_{ck} (°C) is cell temperature at the k th phase; K_v and K_i are the voltage and current temperature coefficients (V/°C) and (A/°C); N_{OT} is the nominal operating temperature of cell (°C) and T_a^t denotes the hourly average ambient temperature (°C).

3. Problem articulation

The problem presented in this paper is to address the challenge of optimizing the locations and sizes of RESs (WTs and solar PV), alongside the PF of WT, within the RDG. The goal is to achieve optimal grid operation through both separate and simultaneous optimizations. The detailed mathematical formulations of the three objective functions and constraints are presented in the next subsections.

3.1. Objective functions

The main goal of this optimization problem is to reduce power losses and the voltage deviation, while improve the voltage stability index while adhering to various equality and inequality constraints.

3.2. Power loss reduction

Radial distribution grids suffer from elevated real power losses due to their high line resistance relative to reactance ratios. Evaluating and reducing these losses is crucial to enhance the operational efficiency of the grid. Fig. 2 represents an equivalent two-bus model of the grid. After power flow calculation, the annual average power loss can be expressed as follows,

$$P_{loss} = \frac{\sum_{t=1}^{N_t} \sum_{i=1}^{N_b} R_{bi} \left(\frac{V_i^t - V_{i+1}^t}{R_{bi} + jX_{bi}} \right)^2}{N_t} \quad (19)$$

where, V_i and V_{i+1} represent the voltage magnitude at the sending bus i and receiving bus $i + 1$ for the t th hour, respectively. R_{bi} and X_{bi} are respectively the resistance and reactance of the i th branch; N_b denotes the total number of branches and N_t denotes the total number of time periods (hours).

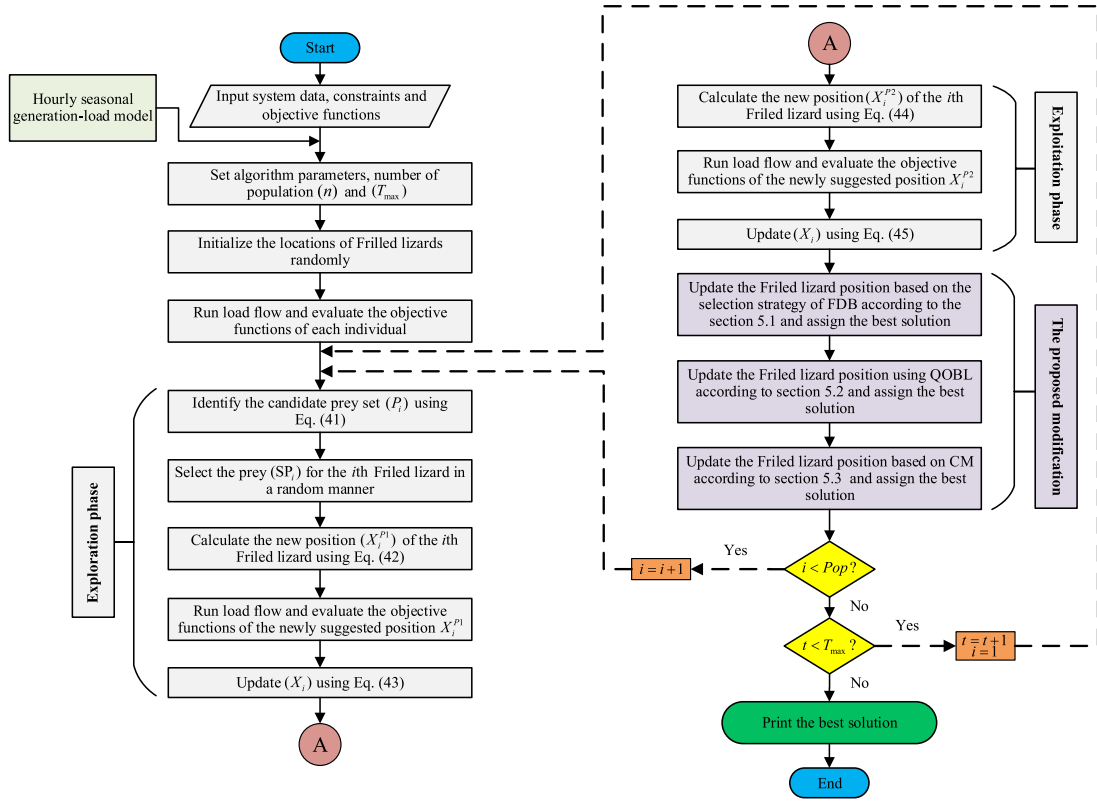


Fig. 3. Simulation flowchart for the proposed method.

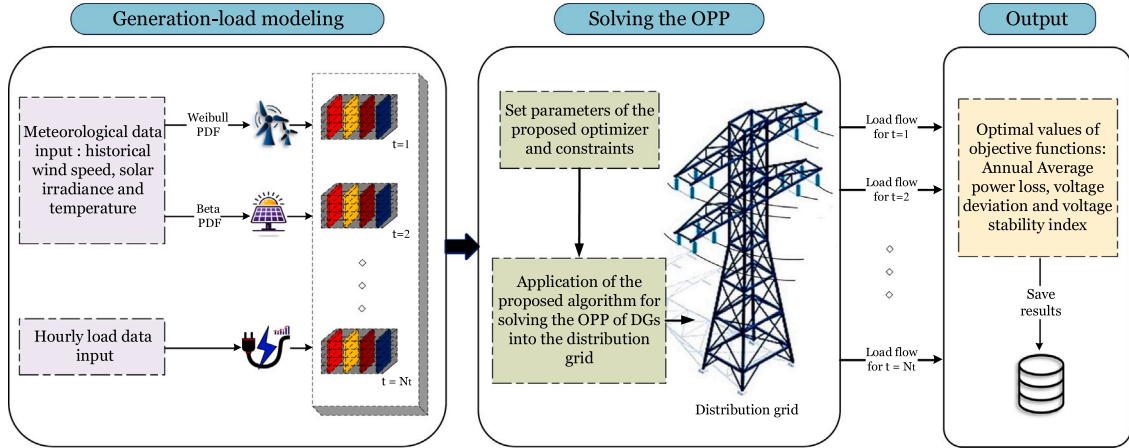


Fig. 4. Simulation flow process for addressing the OPP of RESs into the distribution grid.

3.2.1. Voltage deviation minimization

Voltage deviation (VD) refers to the difference in voltage magnitude at various bus points from a specified reference value (V_{ref}) equal to 1 p.u. Minimizing VD is crucial for meeting operational requirements and maintaining high service quality standards. The annual average total VD can be formulated as follows,

$$VD = \frac{\sum_{t=1}^{N_t} \sum_{i=1}^N |V_{ref} - |V_i^t||^2}{N_t} \quad (20)$$

where, N denotes the total number of buses, and V_i^t refers to the voltage magnitude at the bus i for the t th hour.

3.2.2. Voltage stability index maximization

The Voltage Stability Index (VSI) reflects the power system's capability to maintain steady voltage levels. For stable grid operation, the

VSI should be greater than zero across all the buses of a distribution grid. Buses with lower VSI values are more susceptible to voltage instability. Therefore VSI must be maximized to insure system stability. The mathematical expression of the VSI at the $(i+1)$ th bus for a radial distribution grid at the t th hour is given as follows [39,40]:

$$VSI_{i+1}^t = |V_i^t|^4 - 4(P_{di+1}^t R_{bi} + Q_{di+1}^t X_{bi})|V_i^t|^2 - 4(P_{di+1}^t X_{bi} - Q_{di+1}^t R_{bi})^2 \quad (21)$$

where, P_{di+1}^t and Q_{di+1}^t are respectively the demand for active and reactive powers at the $(i+1)$ th bus for the t th hour. The annual average of voltage stability for the entire distribution grid can be expressed as,

$$VSI = \frac{\sum_{t=1}^{N_t} \sum_{i=1}^{N-1} VSI_{i+1}^t}{N_t} \quad (22)$$

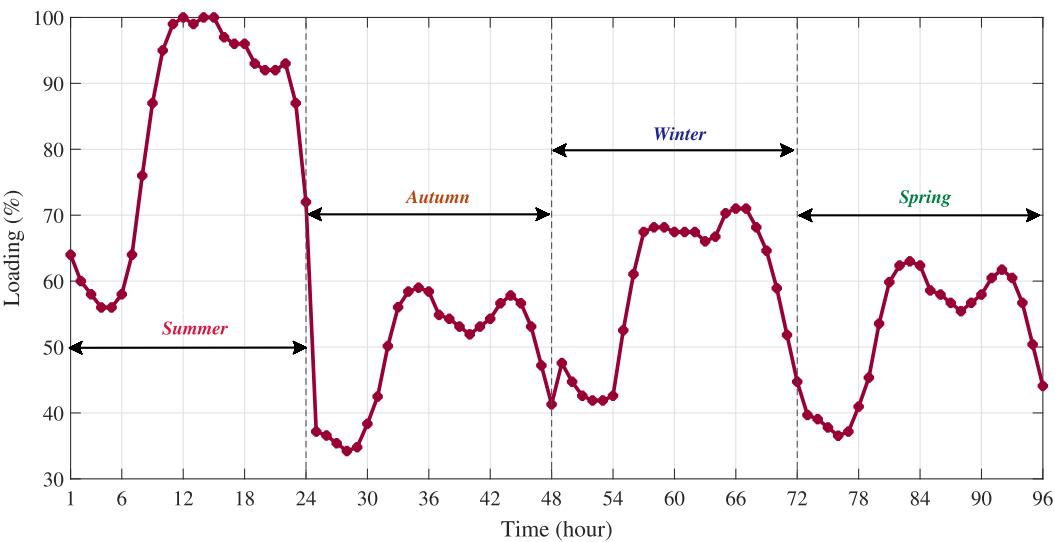


Fig. 5. Seasonal hourly load profile.



Fig. 6. The studied area location on Google maps.

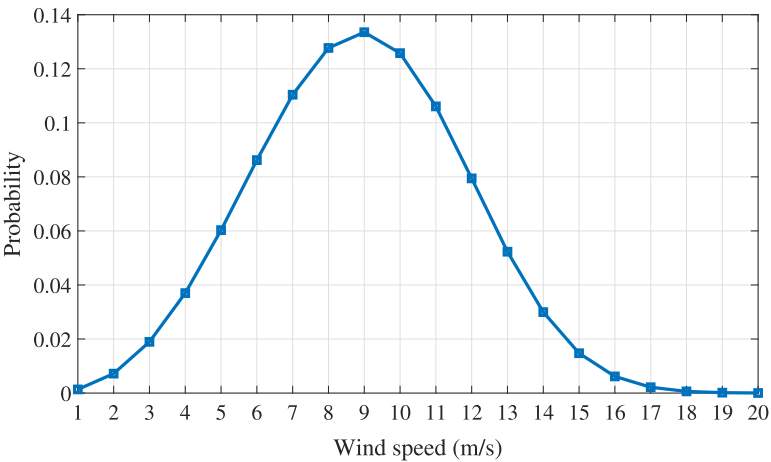


Fig. 7. Probability distribution of wind speed for 22th hour of summer ($k=3.4217$, $c=9.8850$).

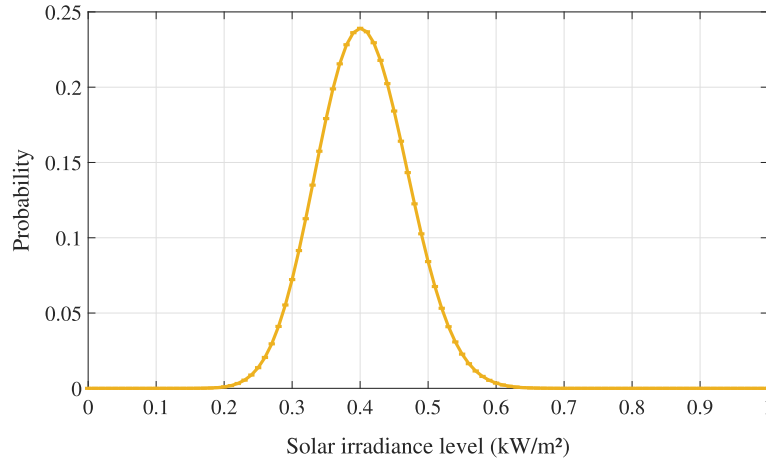


Fig. 8. Beta distribution of solar irradiance for 14th hour of spring ($\mu=0.4042$, $\sigma=0.1012$).

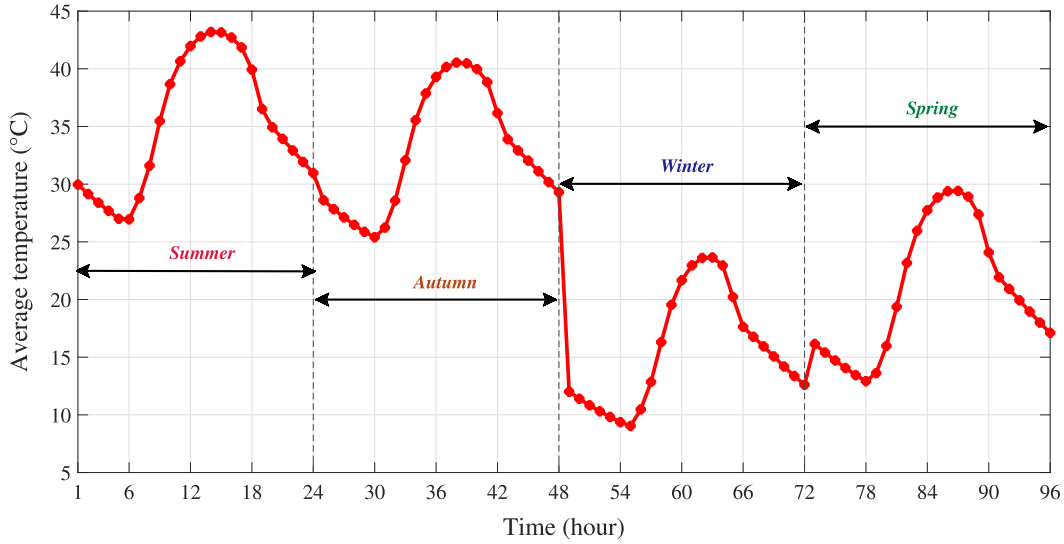


Fig. 9. Seasonal hourly average temperature.

3.2.3. The multi objective function

The optimization problem considers the aforementioned objective functions simultaneously (power loss, voltage deviation and voltage stability index). The weighted aggregation approach is used to formulate the multi-objective function. Therefore, the fitness function can be described in the following way,

$$Fit = \delta_1 \times F_1 + \delta_2 \times F_2 + \delta_3 \times F_3 \quad (23)$$

with,

$$\sum_{i=1}^3 \delta_i = 1, \delta_i \in [0, 1] \quad (24)$$

The following formulas are for the normalization of the objective functions

$$F_1 = \frac{P_{loss}}{P_{loss_{base}}} \quad (25)$$

$$F_2 = \frac{VD}{VD_{base}} \quad (26)$$

$$F_3 = \frac{VSI}{VSI_{base}} \quad (27)$$

where, F_1 , F_2 , and F_3 denote the normalized objective functions, with δ_1 , δ_2 , and δ_3 being the weighting factors are selected as 0.5, 0.25, and 0.25, respectively [41].

3.3. System limitations

In addressing the OPP of RESs, the proposed planning technique completely meets all system limitations, including the equality and inequality constraints, as highlighted below.

3.3.1. Equality constraints

Maintaining a balance between power generation and the total power demand plus losses is essential to prevent reverse power flow, which could damage the system. These balance constraints are expressed as follows:

$$P_{sub}^t + \sum_{i=1}^{N_f} P_{WT,i}^t \times n_{WT,i} + \sum_{i=1}^{N_f} P_{PVS,i}^t \times n_{PVS,i} = \sum_{j=2}^N P_{d,j}^t + \sum_{j=1}^{Nb} P_{loss,j}^t \quad (28)$$

$$Q_{sub}^t + \sum_{i=1}^{N_f} Q_{WT,i}^t \times n_{WT,i} = \sum_{j=2}^N Q_{d,j}^t + \sum_{j=1}^{Nb} Q_{loss,j}^t \quad (29)$$

Table 2

The statistical analysis of diverse optimization algorithms on the standard functions.

Function	Measure	GWO [27]	HHO [28]	CDO [29]	DOA [30]	WOA [31]	BKA [32]	FLO [33]	IFLO
F1	Best	1.287E-16	3.766E-75	1.892E-82	4.120E-247	2.197E-54	6.818E-63	5.179E-245	0
	Average	3.442E-15	1.767E-60	8.327E-79	2.167E-28	3.221E-42	7.522E-44	6.963E-227	0
	Worst	1.858E-14	5.218E-59	8.524E-78	6.500E-27	4.059E-41	2.256E-42	2.022E-225	0
	SD	3.834E-15	9.522E-60	1.912E-78	1.187E-27	9.726E-42	4.120E-43	0	0
F2	Best	5.297E-10	2.489E-38	1.189E-43	1.376E-124	3.446E-35	3.106E-31	1.549E-122	0
	Average	1.316E-09	7.004E-33	6.960E-41	3.737E-23	9.276E-30	1.741E-28	7.132E-114	0
	Worst	3.509E-09	7.354E-32	4.876E-40	1.111E-21	1.942E-28	1.143E-27	2.118E-112	0
	SD	7.725E-10	1.650E-32	1.070E-40	2.028E-22	3.561E-29	3.153E-28	3.865E-113	0
F3	Best	5.384E-04	1.232E-63	6.503E-71	2.714E-289	2.866E+04	2.358E-62	3.595E-212	0
	Average	8.805E-02	1.089E-45	3.142E-54	8.763E-36	6.282E+04	3.884E-49	1.374E-162	0
	Worst	4.287E-01	2.028E-44	9.421E-53	2.629E-34	1.089E+05	7.588E-48	4.121E-161	0
	SD	1.086E-01	4.235E-45	1.720E-53	4.800E-35	1.963E+04	1.549E-48	7.372E-162	0
F4	Best	2.486E-04	2.575E-37	7.006E-40	2.368E-151	8.811E-02	2.378E-31	4.299E-125	0
	Average	6.748E-04	5.684E-31	1.267E-37	6.268E-25	5.360E+01	4.300E-24	7.956E-117	0
	Worst	1.494E-03	1.580E-29	7.342E-37	1.331E-23	8.827E+01	1.289E-22	1.236E-115	0
	SD	3.103E-04	2.880E-30	1.861E-37	2.562E-24	3.300E+01	2.353E-23	3.019E-116	0
F5	Best	2.593E+01	1.471E-04	2.789E+01	2.882E+01	2.787E+01	2.733E+01	0	0
	Average	2.747E+01	3.370E-02	2.819E+01	2.892E+01	2.841E+01	2.826E+01	0	0
	Worst	2.877E+01	3.283E-01	2.874E+01	2.899E+01	2.881E+01	2.896E+01	0	0
	SD	8.984E-01	6.005E-02	2.243E-01	4.198E-02	3.590E-01	4.958E-01	0	0
F6	Best	2.234E-01	7.151E-07	7.500E+00	3.497E+00	1.753E-01	1.554E+00	0	0
	Average	9.570E-01	4.801E-04	7.500E+00	5.636E+00	8.839E-01	3.049E+00	0	0
	Worst	1.996E+00	3.278E-03	7.500E+00	6.898E+00	1.690E+00	6.414E+00	0	0
	SD	4.891E-01	7.050E-04	0	8.675E-01	4.102E-01	1.224E+00	0	0
F7	Best	1.639E-03	2.108E-05	8.888E-06	3.961E-05	2.851E-05	8.786E-05	5.163E-06	5.354E-07
	Average	3.711E-03	2.981E-04	1.630E-04	8.825E-04	5.719E-03	5.436E-04	8.908E-05	5.209E-05
	Worst	9.103E-03	2.256E-03	7.011E-04	2.949E-03	2.886E-02	1.413E-03	3.692E-04	1.438E-04
	SD	1.653E-03	4.127E-04	1.401E-04	9.445E-04	6.123E-03	3.324E-04	7.645E-05	3.517E-05
F8	Best	-8.063E+03	-1.257E+04	-5.167E+03	-6.554E+03	-1.256E+04	-1.050E+04	-1.257E+04	-1.257E+04
	Average	-6.010E+03	-1.256E+04	-3.728E+03	-4.724E+03	-9.889E+03	-8.817E+03	-9.597E+03	-1.079E+04
	Worst	-3.426E+03	-1.235E+04	-2.936E+03	-3.455E+03	-6.684E+03	-7.081E+03	-8.681E+03	-9.016E+03
	SD	1.028E+03	4.002E+01	4.744E+02	7.277E+02	1.912E+03	8.633E+02	1.353E+03	1.807E+03
F9	Best	4.093E-12	0	0	0	0	0	0	0
	Average	8.162E+00	0	1.088E+02	0	0	0	0	0
	Worst	2.349E+01	0	2.512E+02	0	0	0	0	0
	SD	6.094E+00	0	1.149E+02	0	0	0	0	0
F10	Best	4.975E-09	8.882E-16	4.441E-15	8.882E-16	4.441E-15	8.882E-16	8.882E-16	8.882E-16
	Average	1.156E-08	8.882E-16	4.441E-15	1.007E-15	7.046E-15	1.007E-15	8.882E-16	8.882E-16
	Worst	3.124E-08	8.882E-16	4.441E-15	4.441E-15	1.510E-14	4.441E-15	8.882E-16	8.882E-16
	SD	5.925E-09	0	0	6.486E-16	2.273E-15	6.486E-16	0	0
F11	Best	2.554E-15	0	0	0	0	0	0	0
	Average	6.080E-03	0	3.404E-03	0	5.407E-02	0	0	0
	Worst	3.807E-02	0	2.346E-02	0	5.527E-01	0	0	0
	SD	1.099E-02	0	6.600E-03	0	1.470E-01	0	0	0
F12	Best	2.042E-02	6.700E-08	1.106E+00	2.512E-01	1.570E-02	6.368E-02	1.571E-32	1.571E-32
	Average	5.770E-02	2.588E-05	1.451E+00	5.677E-01	5.879E-02	1.890E-01	1.571E-32	1.571E-32
	Worst	1.400E-01	1.245E-04	1.669E+00	1.626E+00	4.630E-01	6.891E-01	1.571E-32	1.571E-32
	SD	3.109E-02	3.245E-05	2.723E-01	2.770E-01	8.221E-02	1.611E-01	5.567E-48	5.567E-48
F13	Best	4.533E-01	1.323E-06	3.547E-01	2.119E+00	2.817E-01	1.188E+00	1.350E-32	1.350E-32
	Average	8.546E-01	2.315E-04	5.825E-01	2.861E+00	7.618E-01	2.090E+00	1.350E-32	1.350E-32
	Worst	1.479E+00	8.604E-04	8.250E-01	3.156E+00	1.442E+00	2.998E+00	1.350E-32	1.350E-32
	SD	2.540E-01	2.270E-04	1.146E-01	2.634E-01	3.073E-01	4.718E-01	5.567E-48	5.567E-48
F14	Best	9.980E-01	9.980E-01	8.937E+00	9.980E-01	9.980E-01	9.980E-01	9.980E-01	9.980E-01
	Average	5.761E+00	1.592E+00	1.506E+01	2.670E+00	2.837E+00	1.295E+00	9.982E-01	9.980E-01
	Worst	1.267E+01	5.929E+00	1.830E+01	5.929E+00	1.076E+01	4.950E+00	1.003E-00	9.980E-01
	SD	4.386E+00	1.285E+00	2.611E+00	1.766E+00	2.618E+00	8.677E-01	9.639E-04	0
F15	Best	3.160E-04	3.081E-04	3.164E-04	3.075E-04	3.086E-04	3.075E-04	3.196E-04	3.075E-04
	Average	5.806E-03	4.139E-04	3.740E-04	2.918E-03	8.229E-04	2.492E-03	7.698E-04	3.111E-04
	Worst	2.036E-02	1.630E-03	7.761E-04	2.255E-02	2.252E-03	2.036E-02	1.661E-03	3.266E-04
	SD	8.930E-03	2.357E-04	9.749E-05	6.435E-03	6.231E-04	6.071E-03	4.607E-04	4.821E-06
F16	Best	-1.032E+00	-1.032E+00	-1.029E+00	-1.032E+00	-1.032E+00	-1.032E+00	-1.032E+00	-1.032E+00
	Average	-1.032E+00	-1.032E+00	-1.004E+00	-1.032E+00	-1.032E+00	-1.032E+00	-1.029E+00	-1.032E+00
	Worst	-1.032E+00	-1.032E+00	-9.999E-01	-1.032E+00	-1.032E+00	-1.032E+00	-1.012E+00	-1.032E+00
	SD	6.438E-08	6.040E-08	8.734E-03	1.803E-10	8.293E-09	5.455E-16	5.022E-03	6.584E-16

(continued on next page)

Table 2 (continued).

Function	Measure	GWO [27]	HHO [28]	CDO [29]	DOA [30]	WOA [31]	BKA [32]	FLO [33]	IFLO
F17	Best	3.979E-01	3.979E-01	3.979E-01	3.979E-01	3.979E-01	3.979E-01	3.979E-01	3.979E-01
	Average	3.980E-01	3.980E-01	4.058E-01	3.979E-01	3.979E-01	3.979E-01	4.216E-01	3.979E-01
	Worst	3.996E-01	3.987E-01	5.820E-01	3.979E-01	3.981E-01	3.979E-01	6.257E-01	3.979E-01
	SD	3.115E-04	1.690E-04	3.334E-02	3.243E-16	4.157E-05	0	5.690E-02	0
F18	Best	3.000E+00	3.000E+00	3.000E+00	3.000E+00	3.000E+00	3.000E+00	3.000E+00	3.000E+00
	Average	3.000E+00	3.000E+00	3.182E+01	3.000E+00	3.000E+00	3.000E+00	9.163E+00	3.000E+00
	Worst	3.001E+00	3.000E+00	8.404E+01	3.000E+00	3.004E+00	3.000E+00	3.131E+01	3.000E+00
	SD	1.192E-04	5.150E-06	3.510E+01	2.125E-07	6.823E-04	3.141E-15	9.370E+00	1.330E-15
F19	Best	-3.863E+00	-3.863E+00	-3.862E+00	-3.863E+00	-3.863E+00	-3.863E+00	-3.862E+00	-3.863E+00
	Average	-3.862E+00	-3.858E+00	-3.858E+00	-3.828E+00	-3.852E+00	-3.863E+00	-3.705E+00	-3.863E+00
	Worst	-3.860E+00	-3.852E+00	-3.848E+00	-3.090E+00	-3.810E+00	-3.863E+00	-2.907E+00	-3.863E+00
	SD	7.728E-04	3.401E-03	3.347E-03	1.439E-01	1.323E-02	4.025E-13	1.814E-01	2.710E-15
F20	Best	-3.322E+00	-3.277E+00	-3.314E+00	-3.322E+00	-3.321E+00	-3.322E+00	-3.128E+00	-3.322E+00
	Average	-3.271E+00	-3.011E+00	-3.219E+00	-3.185E+00	-3.238E+00	-3.293E+00	-2.629E+00	-3.290E+00
	Worst	-3.084E+00	-2.600E+00	-3.101E+00	-2.578E+00	-3.031E+00	-3.202E+00	-1.953E+00	-3.203E+00
	SD	7.157E-02	1.596E-01	5.720E-02	1.473E-01	1.006E-01	5.082E-02	3.297E-01	5.348E-02
F21	Best	-1.015E+01	-9.825E+00	-9.627E+00	-1.015E+01	-1.015E+01	-1.015E+01	-1.015E+01	-1.015E+01
	Average	-8.212E+00	-5.201E+00	-5.374E+00	-7.597E+00	-9.652E+00	-9.652E+00	-9.139E+00	-1.015E+01
	Worst	-2.369E+00	-4.996E+00	-2.795E+00	-2.630E+00	-2.627E+00	-2.630E+00	-5.055E+00	-1.015E+01
	SD	2.867E+00	8.734E-01	1.510E+00	2.977E+00	2.594E+00	1.909E+00	1.915E+00	6.019E-15
F22	Best	-1.040E+01	-5.088E+00	-6.926E+00	-1.040E+01	-1.040E+01	-1.040E+01	-1.040E+01	-1.040E+01
	Average	-1.040E+01	-5.077E+00	-5.240E+00	-7.430E+00	-7.817E+00	-9.958E+00	-9.741E+00	-1.040E+01
	Worst	-1.039E+01	-5.029E+00	-3.209E+00	-2.766E+00	-1.836E+00	-3.724E+00	-5.088E+00	-1.040E+01
	SD	2.244E-03	1.405E-02	9.806E-01	3.313E+00	3.169E+00	1.694E+00	1.679E+00	4.665E-16
F23	Best	-1.054E+01	-5.128E+00	-7.510E+00	-1.054E+01	-1.053E+01	-1.054E+01	-1.054E+01	-1.054E+01
	Average	-9.992E+00	-5.037E+00	-5.503E+00	-7.595E+00	-5.368E+00	-9.866E+00	-9.82E+00	-1.054E+01
	Worst	-2.422E+00	-2.764E+00	-2.675E+00	-2.419E+00	-1.676E+00	-3.833E+00	-5.128E+00	-1.054E+01
	SD	2.058E+00	4.295E-01	1.218E+00	3.476E+00	2.850E+00	2.045E+00	1.782E+00	1.143E-15

where P_{sub}^t and Q_{sub}^t are respectively the total active and reactive power supplied by the substation into the distribution grid at the t th hour. P_{WT}^t and P_{PVS}^t are respectively the power generated by WT and solar PV string at the t th hour. N_f symbolizes the number of farms, while $n_{WT,i}$ and $n_{PV,i}$ signify the number of installed units in each WT and solar PV farm. $P_{d,j}^t$ and $Q_{d,j}^t$ represent the active and reactive power demands for a branch j at a time period t . The total active and reactive system losses at branch j at a time period t are represented using $P_{loss,j}^t$ and $Q_{loss,j}^t$, respectively.

3.3.2. Inequality constraints

- DG location limit: Each DG type can be installed on any bus throughout the system, except for the slack bus (substation), as follows:

$$2 \leq DG_{loc,j} \leq N \quad j \in type \quad (30)$$

- Bus voltage limit: The voltage magnitude at each bus for every hour t must kept within the specified standard limits, as follows:

$$V_{min} \leq V_i^t \leq V_{max} \quad (31)$$

In the inequality above, V_{min} and V_{max} denote the minimum and maximum permissible voltage levels for any bus in each hour.

- DG penetration limits: The combined rated capacity of RESs must not surpass the overall load demand, as described below.

$$\sum_{i=1}^{N_f} P_{WT, rated}^t \times n_{WT,i} + \sum_{i=1}^{N_f} P_{PVS, rated}^t \times n_{PVS,i} \leq \sum_{j=2}^N P_{d,j}^t \quad (32)$$

- Power factor limit : The PF of each wind farm is treated as an adjustable parameter within the optimization process, as it allows for reactive power support when required. It is constrained within the following range:

$$PF_{min} \leq PF_i \leq PF_{max} \quad i = 1, 2, \dots, N_f \quad (33)$$

Thus, the corresponding reactive power supplied by each wind farm at each hour ' t ' is expressed as a function of active power

and PF as follows:

$$Q_{WT}^t \times n_{WT,i} = P_{WT}^t \times n_{WT,i} \times \tan(\cos^{-1}(PF_i)) \quad i = 1, 2, \dots, N_f \quad (34)$$

In contrast, PV systems are assumed to operate at a unity power factor (PF = 1), as they are typically configured to generate only active power without injecting or absorbing reactive power.

- Thermal limit : To maintain line security, the current flowing through branch i at the t th hour, symbolized as I_{bi}^t , must remain below its specified maximum limit, as follows:

$$I_{bi}^t \leq I_{bi}^{max} \quad i = 1, 2, \dots, N_b \quad (35)$$

where I_{bi}^{max} signifies the maximum current carrying capacity of the i th branch.

4. Frilled Lizard Optimization algorithm

The FLO is a bio-inspired algorithm that simulates the frilled lizard's hunting strategy. This lizard waits patiently for prey, then swiftly attacks and feeds on it before retreating up a tree. FLO mimics this behavior by exploring the solution space, rapidly refining candidate solutions, and strategically converging to optimal results [33].

4.1. Initialization

FLO begins by creating a set of random solutions within the search space, where each frilled lizard represents a potential solution and is formulated as a vector, initialized using Eq. (35). Collectively, the FLO population can be expressed in the form of a matrix X , as shown in Eq. (36). During each iteration of the optimization process, the algorithm evaluates the fitness of each solution and identifies the best candidate among them.

$$x_{i,k} = Lb + rand \times (Ub - Lb) \quad k = 1, 2, \dots, dim. \quad (36)$$

$$X = \begin{bmatrix} X_1 \\ \vdots \\ X_i \\ \vdots \\ X_n \end{bmatrix} = \begin{bmatrix} x_{1,1} & \cdots & x_{1,k} & \cdots & x_{1,dim} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ x_{i,1} & \cdots & x_{i,k} & \cdots & x_{i,dim} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ x_{n,1} & \cdots & x_{n,k} & \cdots & x_{n,dim} \end{bmatrix} \quad (37)$$

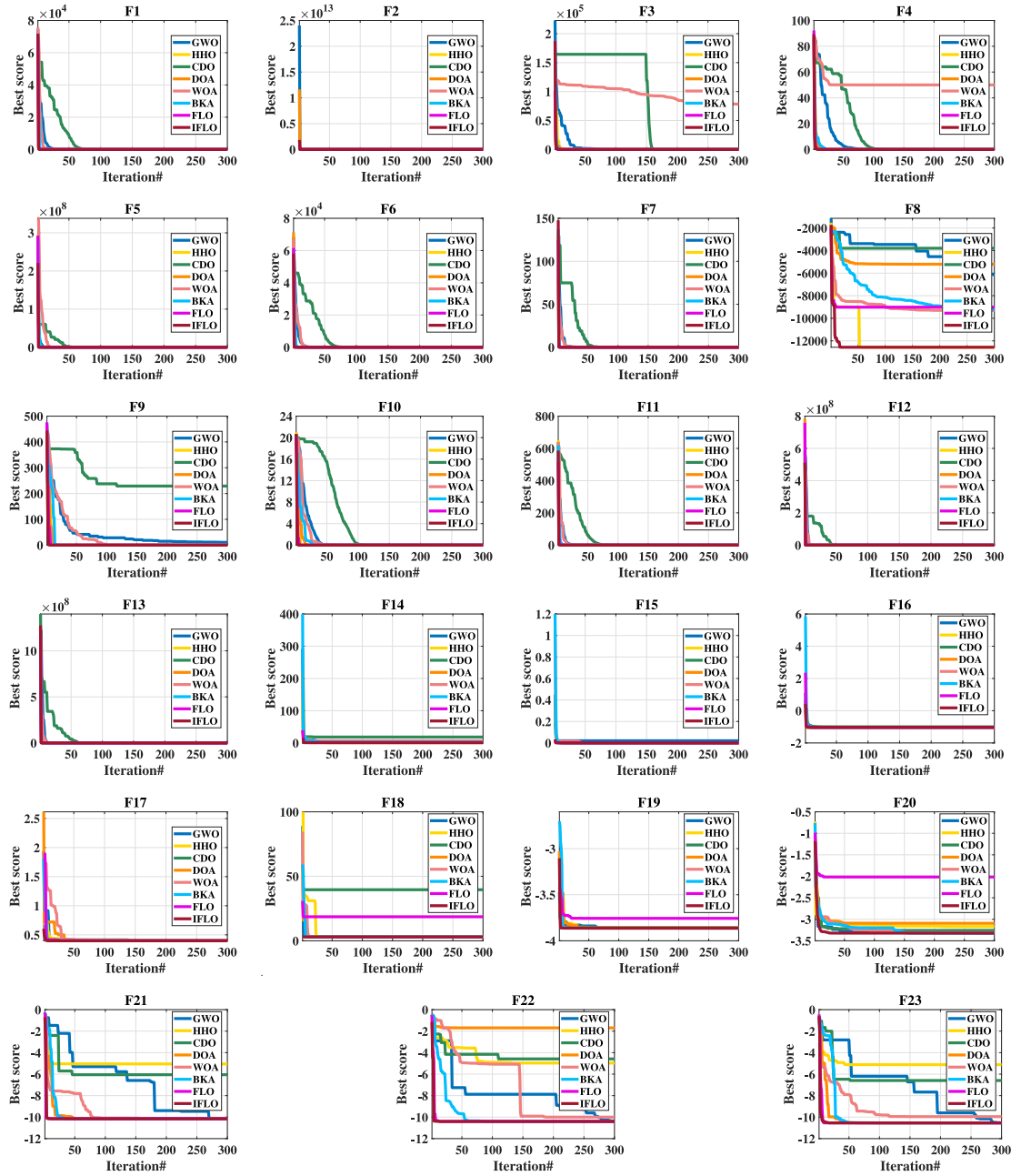


Fig. 10. Convergence curves comparison.

Here, X represents the FLO population matrix, while X_i denotes the i th candidate solution. $x_{i,k}$ represents the k th decision variable at the i th candidate solution, dim refers the problem dimension, and n denotes the total number of frilled lizards.

4.2. Hunting strategy: Exploration phase

One distinctive natural behavior of the frilled lizard is its hunting strategy as a sit-and-wait predator, attacking prey upon sight. Simulating this movement leads to significant shifts in population positions within the problem-solving space, enhancing the algorithm's global search exploration. In the initial phase of FLO, the population positions are updated based on this strategy. Each frilled lizard considers the positions of other members with better fitness values as prey positions, determined by Eq. (37) [33].

$$P_i = \{X_j : F_j < F_i \text{ and } j \neq i\}, \quad i = 1, 2, \dots, n \text{ and } j \in \{1, 2, \dots, n\} \quad (38)$$

Here, P denotes the group of potential prey for the i th frilled lizard, where X_j represents a member of the population with a better fitness value compared to the i th frilled lizard, and F_j represents the corresponding objective function value.

The FLO design assumes that the frilled lizard randomly selects one of the potential prey and then attacks it. Using the modeled movement of the lizard toward the chosen prey, a new position for each member of the population is determined based on Eq. (38). If this updated position results in a better fitness value, it replaces the previous position of the corresponding individual as per Eq. (39) [33]:

$$x_{i,k}^{P1} = x_{i,k} + rand \times (SP_{i,k} - I \cdot x_{i,k}), \quad i = 1, 2, \dots, n, \text{ and } k = 1, 2, \dots, dim \quad (39)$$

$$X_i = \begin{cases} X_i^{P1}, & F_i^{P1} < F_i \\ X_i, & \text{else} \end{cases} \quad (40)$$

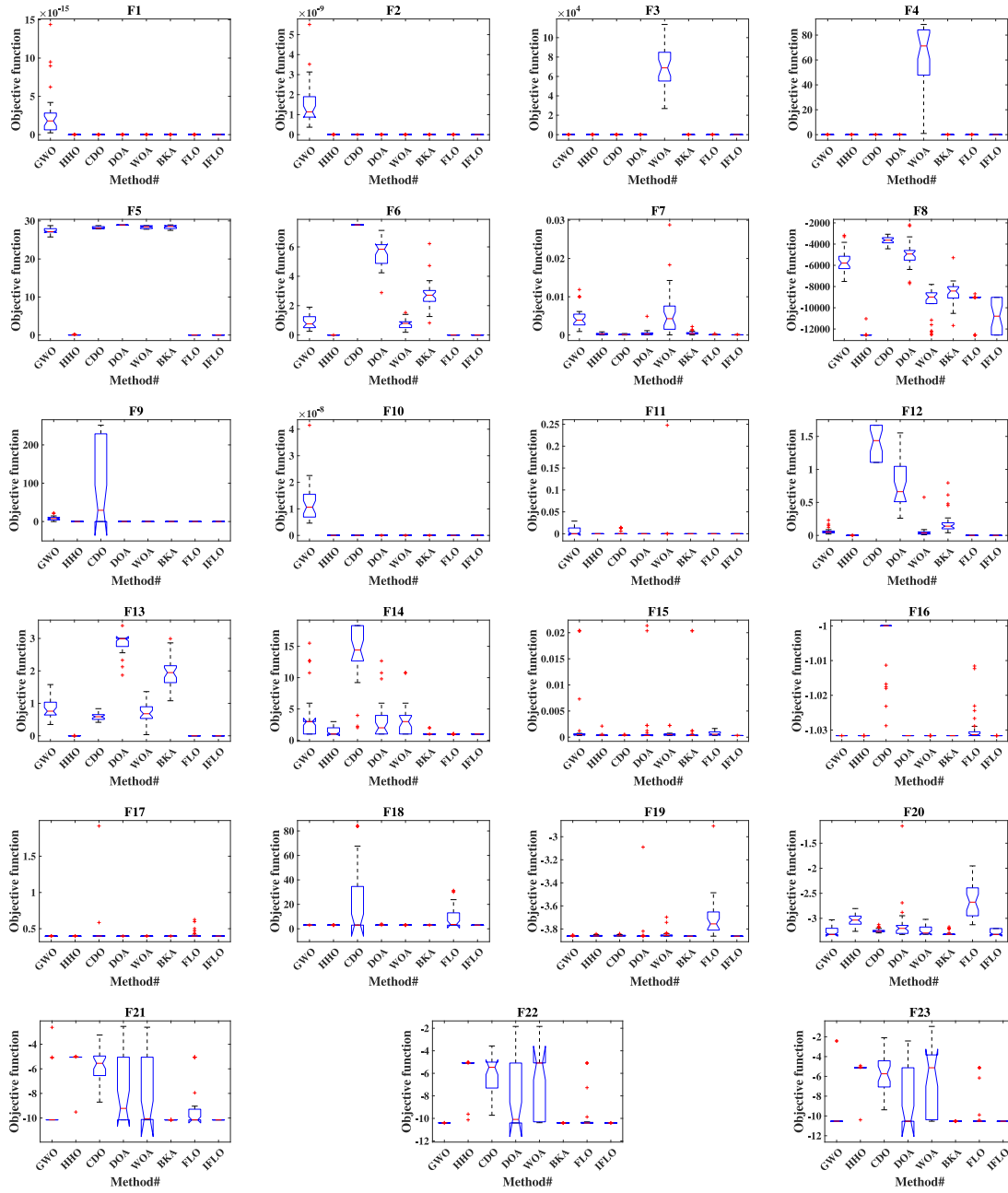


Fig. 11. Boxplots comparison.

where, X_i^{P1} symbolizes the newly proposed position of the i th frilled lizard based on the initial phase of FLO. The term $x_{i,k}^{P1}$ represents its k th dimension, while F_i^{P1} indicates its objective function value. $SP_{i,k}$ is the k th dimension of the chosen prey for the i th frilled lizard, and I is a random number from the set $\{1, 2\}$.

4.3. Moving up the tree: Exploitation phase

Once it finishes feeding, the frilled lizard ascends to the top of a nearby tree. This behavior, when simulated, leads to slight adjustments in the individuals' positions within the search area, thereby boosting the local search capabilities of the algorithm. During the second phase

of the FLO algorithm, the positions of individuals are updated according to the frilled lizard's retreat strategy. By modeling the frilled lizard's movement, a new position for each individual is computed according to Eq. (40). If this updated position enhances the objective function, it replaces the old position, as described in Eq. (41) [33].

$$x_{i,k}^{P2} = x_{i,k} + (1 - 2rand) \cdot \frac{(UB_k - LB_k)}{t}, \quad (41)$$

$$i = 1, 2, \dots, n, \quad k = 1, 2, \dots, dim, \quad \text{and} \quad t = 1, 2, \dots, T$$

$$X_i = \begin{cases} X_i^{P2}, & \text{if } F_i^{P2} < F_i \\ X_i, & \text{else} \end{cases} \quad (42)$$

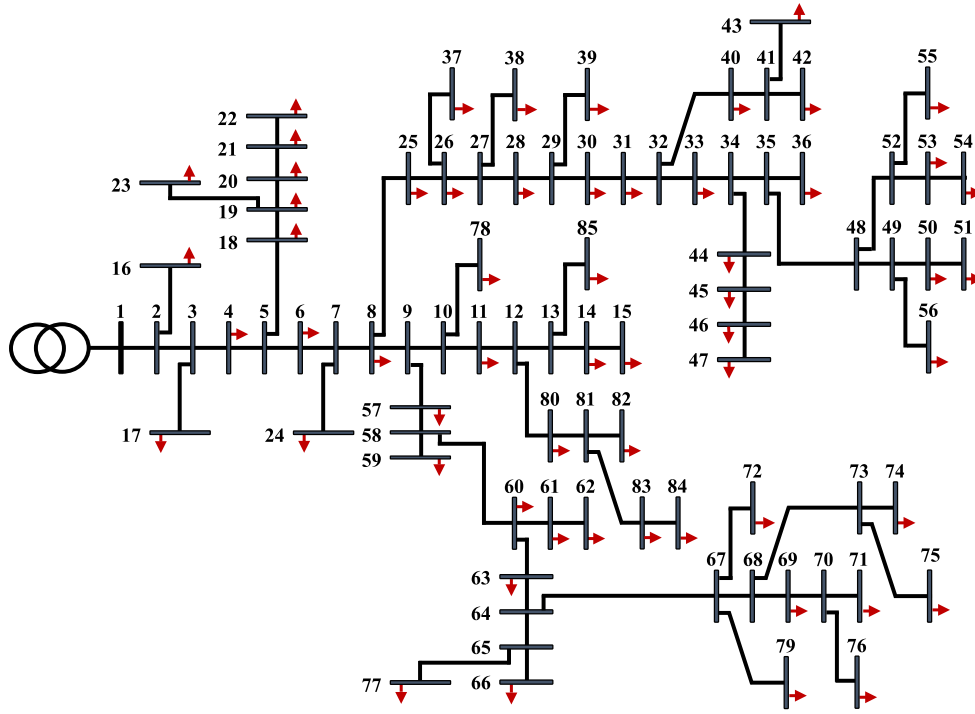


Fig. 12. Schematic diagram of the 85-bus RDG.

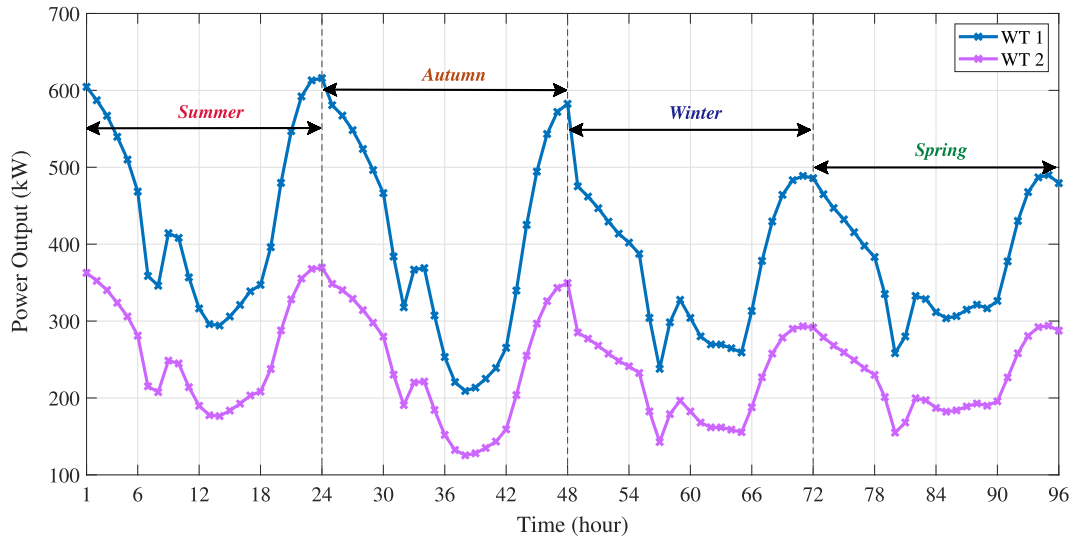


Fig. 13. The seasonal hourly power produced by wind farms.

Here, X_i^{P2} refers to the newly proposed position of the i th frilled lizard according to the second stage of FLO, $x_{i,k}^{P2}$ indicates its d th dimension, F_i^{P2} provides its objective function value, t denotes the current iteration number, and T specifies the maximum number of iterations.

5. Improved Frilled Lizard Optimization algorithm

The suggested Improved Frilled Lizard Optimization (IFLO) algorithm incorporates three distinct strategies: Fitness Distance Balance

(FDB), Quasi-Opposite-Based Learning (QOBL), and Cauchy Mutation (CM), each contributing uniquely to improving the algorithm's performance.

5.1. Fitness distance balance (FDB)

The FDB method is an efficient selection strategy in optimization algorithms that improves the exploration phase by evaluating candidate solutions according to their fitness and proximity to the optimal solution [42–44]. FDB prioritizes solutions with superior fitness values

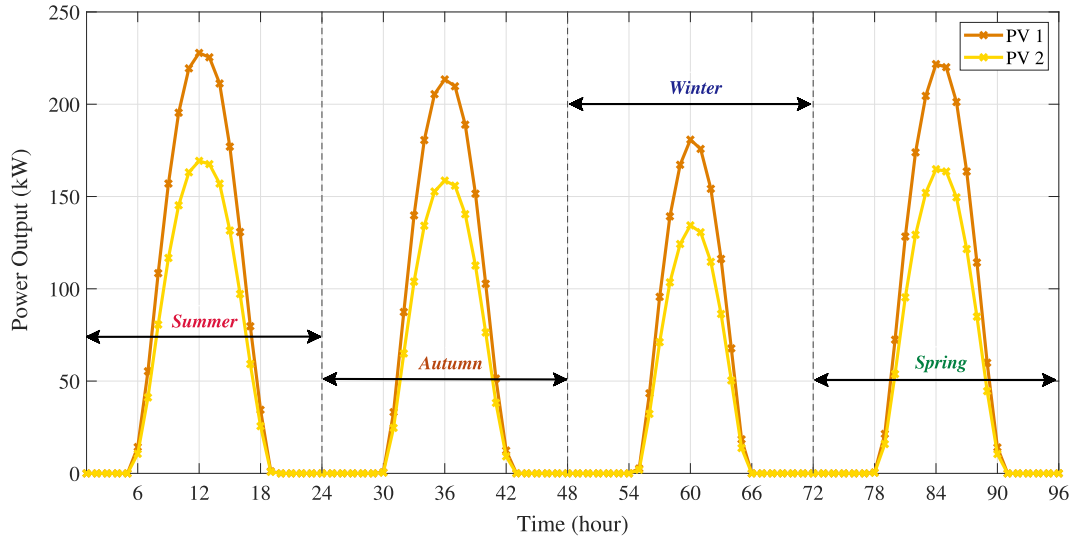


Fig. 14. The seasonal hourly power produced by solar PV farms.

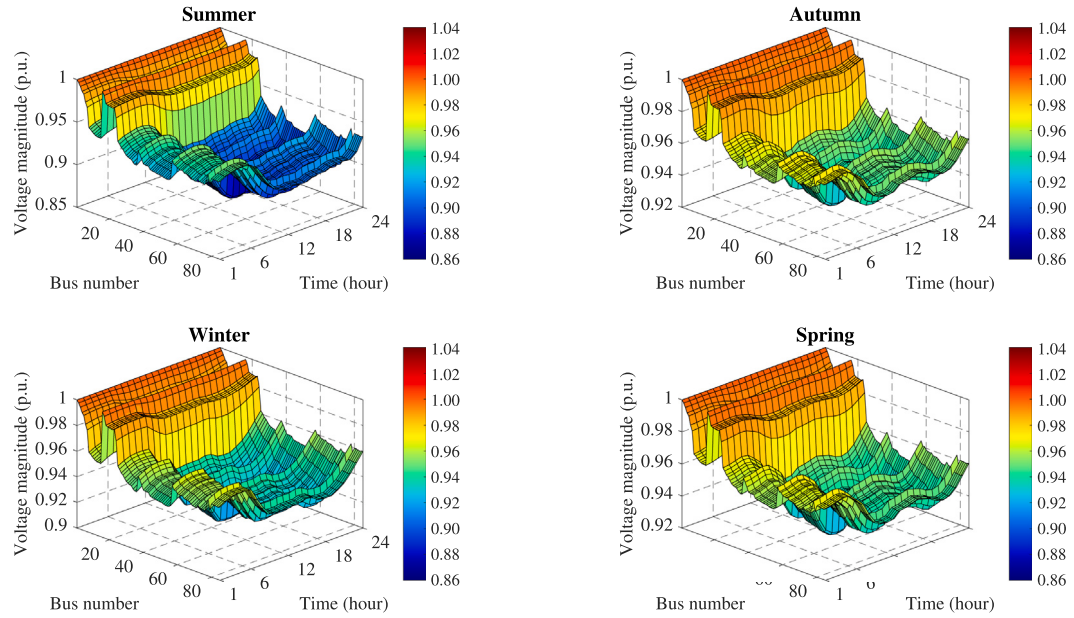


Fig. 15. Seasonal voltage profiles for the 85-bus distribution grid without RESs integration.

and minimal distances to the optimal solution, ensuring the algorithm effectively explores new areas while refining promising regions. The distance of the solution candidates (DP_i) from the best solution (P_{best}) is computed as follows:

$$DP_i = \sqrt{\sum_{k=1}^{dim} (x_{i,k} - P_{best,k})^2}, \quad \forall i \in \{1, 2, \dots, N\}, P_i \neq P_{best} \quad (43)$$

Then, create the vectors for both fitness and distance values in the following way:

$$DP = [DP_1, DP_2, \dots, DP_N] \quad (44)$$

$$F = [F_1, F_2, \dots, F_N] \quad (45)$$

In the last stage of the FDB method, the score is figured using the following formula:

$$S_{FDBi} = \omega \times (1 - normF_i) + (1 - \omega) \times normDP_i \quad (46)$$

where, $normF_i$ and $normDP_i$ are respectively the normalized fitness and distance values. ω is a weight coefficient that also defines the influence of fitness and distance values and can be described as:

$$\omega = 0.5 \times \left(1 + \frac{t}{T}\right) \quad (47)$$

5.2. Quasi-opposite-based learning (QOBL)

QOBL is an advanced strategy derived from Oppositional Based Learning, aiming to improve the exploration and exploitation capabilities of optimization algorithms [44,45]. By evaluating both current solutions and their quasi-opposite counterparts, it explores the search process more effectively, leading to better convergence to the global optimum. In QOBL, the opposite of a candidate solution $X(x_1, x_2, \dots, x_{dim})$, where $x_k \in [LB_k, UB_k] \quad \forall k \in \{1, 2, \dots, dim\}$ is defined as $Y(y_1, y_2, \dots, y_{dim})$. Each opposite point y_k is calculated as $y_k = LB_k + UB_k - x_k$ and

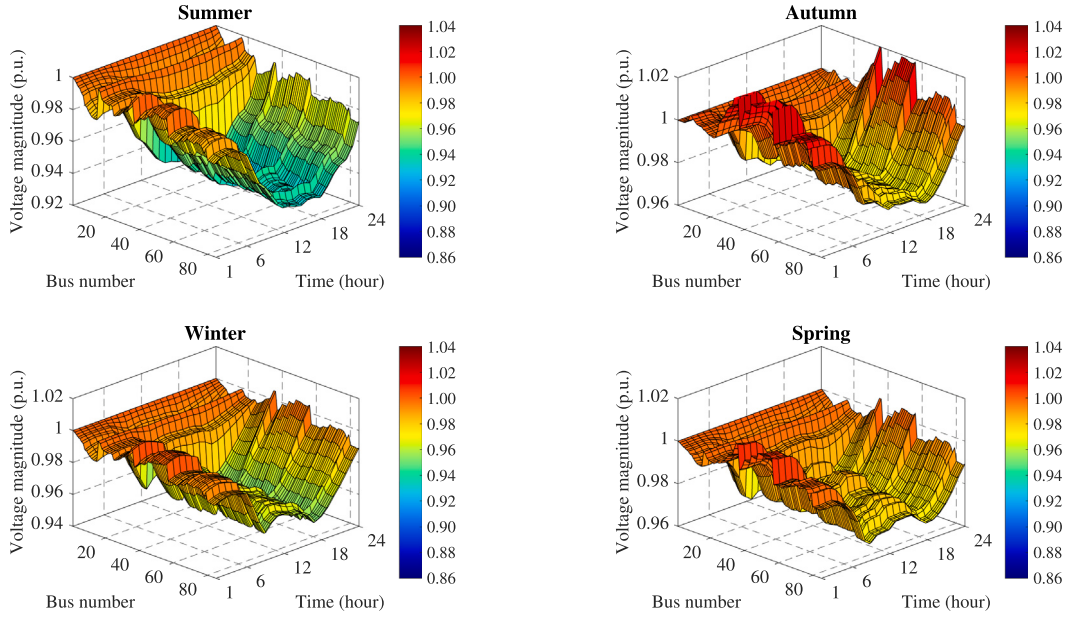


Fig. 16. Seasonal voltage profiles for the 85-bus distribution grid in the presence of RESs.

the midpoint is obtained as :

$$M_k = \frac{LB_k + UB_k}{2} \quad (48)$$

To generate a quasi-opposite point y_k^p , which is more likely to be nearer to the optimal solution than the opposite point y_k , we use the following formula:

$$y_k^q = \begin{cases} M_k + (y_k - M_k) \times rand & \text{if } y_k < M_k \\ y_k + (M_k - y_k) \times rand & \text{otherwise} \end{cases} \quad (49)$$

where $rand$ is the uniformly distributed random number in the range of $[0, 1]$.

5.3. Cauchy mutation (CM)

CM is incorporated to enhance FLO's ability to explore the solution space more thoroughly. This strategy involves injecting significant variability into the population using a CM operator, enabling the algorithm to jump out of the local optima and thereby preventing it from falling into the local optima trap [46–48]. The Cauchy distribution is a continuous probability distribution defined by two parameters (σ, μ). The formula for one-dimensional Cauchy probability density function is presented in Eq. (49):

$$f(z; \sigma, \mu) = \frac{1}{\pi} \frac{\sigma}{\sigma^2 + (z - \mu)^2}, \quad z \in (-\infty, \infty), \quad (50)$$

when $\sigma = 1$ and $\mu = 0$, the Cauchy density function simplifies to its standard form, defined by Eq. (50):

$$f(x; 1, 0) = \frac{1}{\pi(1 + x^2)}, \quad z \in (-\infty, \infty), \quad (51)$$

The formula for the standard Cauchy distribution is represented by Eq. (51):

$$\text{Cauchy}(0, 1) = \tan[\pi(\xi - 0.5)], \quad \xi \in [0, 1]. \quad (52)$$

Eq. (52) is utilized for CM processing on randomly generated new promising positions.

$$X_i^{t+1} = X_i^t + \alpha \otimes \text{Cauchy}(0, 1) \otimes (X_i^t - X_{best}^t) \quad (53)$$

where α denotes to the step size while $\otimes$ represent entrywise multiplication.

Fig. 3 depicts a flowchart of the proposed IFLO algorithm to the OPP solution, while the methodology of the suggested framework is explained in Fig. 4.

6. Data analysis

Analyzing weather data at the specified location, along with load patterns, is crucial as an initial step for solving the OPP of RESs into the distribution grid. The system being studied adheres to the time-varying load curve of the IEEE-RTS system [34,35]. Accordingly, the seasonal hourly load profile is presented in Fig. 5 as a percentage of the annual peak load.

Historical data on wind speed, solar irradiance, and ambient temperature was collected from In Salah region, a province located in central Algeria, in the Sahara, over a decade (from 01 January 2012 to 31 December 2021). The data are available on the NASA website [49]. Fig. 6 shows the precise location of In Salah Province. Each year within this study period is divided into four seasons: summer (May to July), autumn (August to October), winter (November to January), and spring (February to April). Each season is represented by a single day of 24 h, representing the hourly random behavior of RESs during that season. Consequently, four seasonal days represent one year, totaling 96 h (24 h $\times$ 4 seasons). Weibull and beta PDFs are then generated for each hour to accurately fit the hourly seasonal wind speed and solar irradiance, respectively. The adjustment steps are set at 1 m/s for wind speed and 0.05 kW/m² for solar irradiance. For clarification, the probability distribution of wind speed for the 22th hour in summer and solar irradiance for the 14th in spring are represented in Figs. 7 and 8, respectively. The ambient temperature is represented by averaging the hourly temperature values for each season. Thus, the seasonal hourly average temperatures are depicted in Fig. 9.

7. Simulation results and discussions

The suggested IFLO algorithm is designed to solve the OPP of RESs in the 85-bus RDG [50]. Initially, the performance of the proposed algorithm was evaluated through a comparison with several established optimization methods, including the well known GWO [27], HHO [28], CDO [29], DOA [30], WOA [31], BKA [32], and the original FLO [33]. The simulation was conducted using MATLAB R2021a on a computer equipped with an Intel(R) Core(TM) i5-7200U CPU @ 2.50 GHz and

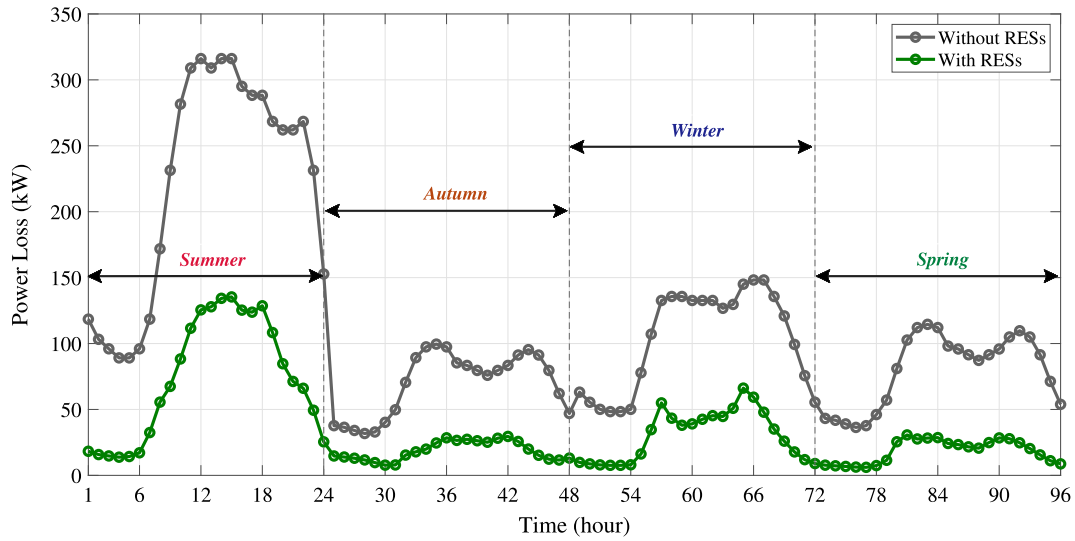


Fig. 17. Seasonal power losses for the 85-bus RDG without and with the integration of RESs.

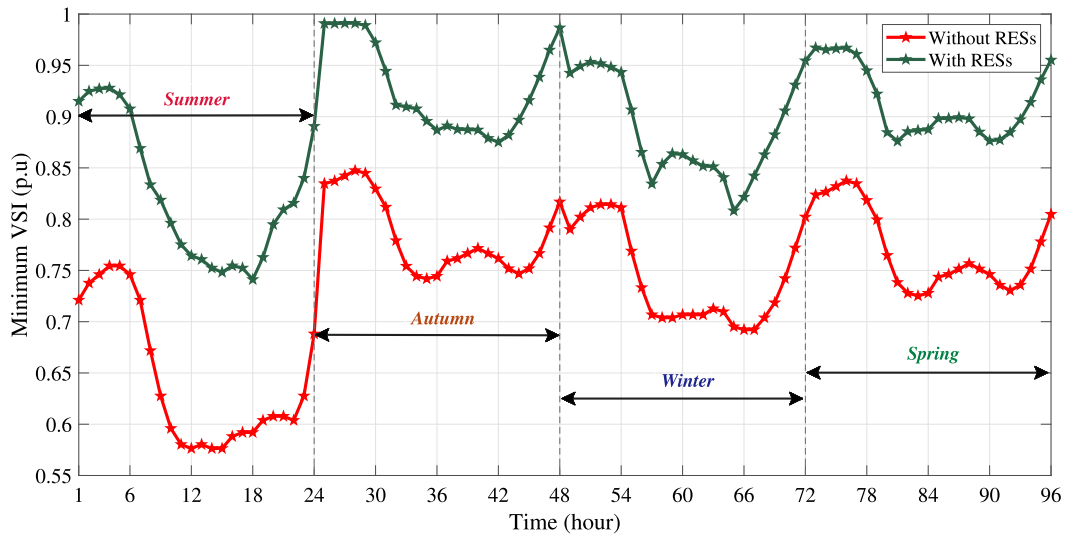


Fig. 18. Minimum VSI recorded at prior and after the integration of RESs for each season and hour.

2.71 GHz, 8.00 GB of RAM, and a 64-bit operating system. The cases covered by the study are detailed as follows:

7.1. Performance assessment of the IFLO algorithm on standard benchmark functions

The proposed IFLO technique was applied to 23 standard benchmark functions. These functions encompass unimodal, multimodal, and fixed-dimension multimodal benchmarks, as detailed in Appendix [51–53]. The parameter settings for the different algorithms are provided in Table 3, and the results were recorded over 30 trial runs.

7.1.1. Analyzing the statistical results

Here, the performance of the suggested IFLO algorithm was compared with other optimization methods, including the GWO [27], HHO [28], CDO [29], DOA [30], WOA [31], BKA [32], and the original FLO [33]. Table 2 presents the statistical results, encompassing the best, average, worst, and standard deviation (SD) of all methods. From the table, it can be observed that the proposed IFLO outperforms other methods in terms of best, average, worst, and SD for the overwhelming majority of the test functions. This confirms that the proposed IFLO

Table 3

Setting parameters of each algorithm.

Algorithm	Parameters for standard benchmark functions
GWO [27]	Population size=30, T_{max} =30, a =[2,0]
HHO [28]	Population size=30, T_{max} =30, E_0 =[-1,1]
CDO [29]	Population size=30, T_{max} =30, r =rand[0,1]
DOA [30]	Population size=30, T_{max} =30, P = 0.5, Q =0.7, β_1 =[-2,2], β_2 =[-1,1]
WOA [31]	Population size=30, T_{max} =30, b =1, l =[-1,1], a_1 = a_2 =[2,0],
BKA [32]	Population size=30, T_{max} =30, p = 0.9
FLO [33]	Population size=30, T_{max} =30, I = rand[1,2]
IFLO	Population size=30, T_{max} =30, I = rand[1,2]

exhibits a higher degree of superiority in finding the best solutions, as compared to the other optimizers.

7.1.2. Convergence curves analysis

The convergence graphs of the proposed IFLO, along with those of other reported techniques, including GWO [27], HHO [28], CDO [29], DOA [30], WOA [31], BKA [32], and the original FLO [33], are

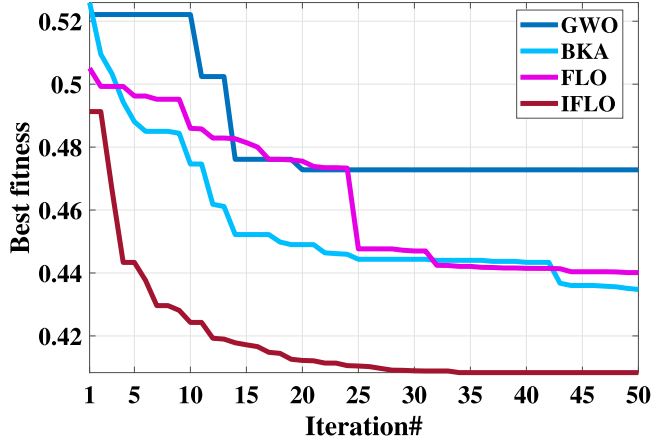


Fig. 19. Convergence curves of different methods addressing the OPP of RESs in the 85-bus RDG.

shown in Fig. 10. According to the convergence curves, the proposed IFLO demonstrates faster convergence for unimodal, multimodal, and fixed-dimension multimodal functions. Furthermore, the proposed IFLO outperforms the conventional FLO, converging to the optimal solution more rapidly. This enhanced performance is due to the proposed modifications, which improve both the exploration and exploitation phases of the optimizer, enabling it to reach the optimal solution more efficiently.

7.1.3. Boxplot analysis

Boxplots are a powerful tool for visualizing data distributions by quartiles, optimally highlighting the key characteristics of the data. In the context of our study, Fig. 11 displays boxplots comparing the performance of the proposed IFLO algorithm against other reported optimizers, namely GWO [27], HHO [28], CDO [29], DOA [30], WOA [31], BKA [32], and the original FLO [33]. These boxplots summarize the distribution of performance across multiple simulation runs using 23 benchmark functions. The narrower boxplot represents the reduced variance in the performance of the IFLO algorithm, meaning it is less prone to outliers and produces results that are more consistent in terms of solution quality and convergence. On the other hand, the wider boxplots for the other algorithms in specific benchmark functions indicate greater variability in performance, suggesting that these algorithms are less consistent in achieving the optimization objectives. The presence of outliers in the compared algorithms boxplots further emphasizes that these methods may occasionally perform poorly in certain scenarios. Hence, the IFLO algorithm outperforms the other methods in terms of consistency, demonstrating its robustness and reliability across multiple simulation scenarios.

7.2. Application of the IFLO for solving the OPP of RESs into the 85-bus distribution grid

In this portion, the suggested IFLO technique is implemented on the 85-bus RDG, employing the developed model serving as input data to address the OPP of RESs. Fig. 12 illustrates the single-line diagram of the IEEE 85-bus RDG of 11 kV. The system comprises 85 buses and 84 branches, having a combined peak load of 2570.28 kW and 2622.08 kVAr. The complete technical descriptions are sourced from [50]. The load flow solution is conducted using the backward/forward sweep method [54]. This research considers the integration of two wind farms and two solar farms into the RDG. The WTs used in this study have a peak capacity of 250 kW. Additionally, each PV string is configured

Table 4

Parameter values specification of PV and WT unit.

Parameters	Value
PV unit	
Current at maximum power point, I_{MPP}	7.76 A
Voltage at maximum power point, V_{MPP}	28.36 V
Short circuit current, I_{sc}	8.38 A
Open circuit voltage, V_{oc}	36.96 V
Nominal cell operating temperature, N_{OT}	49 °C
Current temperature co-efficient, K_i	0.00545 A/°C
Voltage temperature co-efficient, K_v	0.1278 V/°C
WT unit	
Rated output power, P_{rated}	250 kW
Cut-in speed, w_{in}	3 m/s
Rated speed, w_N	12 m/s
Cut-out speed, w_{out}	25 m/s

Table 5

Parameters values limitations.

Parameters	Value
Number of population	30
Maximum iterations	50
Voltage boundaries [24]	$0.9 \text{ p.u.} \leq V_i^t \leq 1.05 \text{ p.u.}$
Size of PV strings	$0 \leq PV \leq 584 \text{ strings}$
Size of WT units	$0 \leq WT \leq 10 \text{ turbines}$
PV system power factor	1
WT Power factor	$0.7 \leq PF \leq 1$

with 20 PV panels, producing a total rated power of 4.4 kW per string. Table 4 offers the characteristics of the PV module and WT [18]. To ensure a fair comparison, the population sizes and the maximum number of iterations were fixed to 30 and 50, respectively, as outlined in Table 5, which also lists the operational system constraints.

The effectiveness of the proposed IFLO algorithm is proved through a comparative study against other recent algorithms, namely GWO, BKA, and the original FLO. These algorithms were also developed using the same dataset, ensuring a fair and consistent comparison in performance across the different methods. The uncertainties related to WTs and solar PVs, along with load fluctuations, are addressed in Section 2 within the context of the developed model through a seasonal planning horizon.

Table 6 presents the energy management results for the 85-bus system obtained using different algorithms before and after the installation of RES units. It provides insights into how WTs and PV strings can enhance system performance. The annual average power losses, voltage deviation, and voltage stability index in the base case without any RESs integrated into the RDG were 118.2953 kW, 0.30662 p.u., and 0.7355 p.u., respectively. Whereas, after the installation of RES units, the proposed IFLO algorithm successfully selects appropriate control parameters for WTs and PV strings. This reduces the annual average power losses from 118.2953 kW to 35.1939 kW (reduction of 70.25%) and the voltage deviation from 0.30662 p.u. to 0.0659 p.u. (improvement of 78.51%), while also enhancing the annual average voltage stability index, rising from 0.7355 p.u. to 0.8884 p.u. (a gain of 20.79%). The optimal locations for WT farms are identified as Bus 35 and Bus 68, with 5 and 3 installed WT units, respectively. Likewise, the optimal locations for PV farms are identified as Bus 52 and Bus 70, with 74 and 55 PV strings, respectively. The corresponding optimal WT ratings are 1250 kW and 750 kW, while the optimal PV ratings are 326 kW and 242 kW, respectively. According to the obtained results in Table 6, it is clear that the proposed IFLO algorithm outperforms other methods, including GWO, BKA, and the original FLO, in achieving optimal results.

Figs. 13 and 14 display the predicted seasonal power generation from each WT and solar PV farm. These figures demonstrate that the power output from WTs and solar PVs varies throughout the year due to

Table 6
Energy management results for OPP of RESs into 85-bus distribution grid.

Factor	Without RESs	With the installation of RESs			
		GWO	BKA	FLO	IFLO
RES Parameters					
Optimal location of WTs	–	63 34	34 68	35 64	35 68
Optimal number of installed WT units	–	1 7	5 3	4 3	5 3
Optimal rated power of WTs (kW)	–	250 1750	1250 750	1000 750	1250 750
Optimal PF of WTs (kW)	–	1 0.7000	0.7068 0.7005	0.7264 0.7066	0.7000 0.7000
Optimal location of PVs	–	46 29	42 51	45 48	52 70
Optimal number of PV strings	–	25 84	114 12	83 103	74 55
Rated power of PVs (kW)	–	110 370	502 53	365 453	326 242
System Performance Metrics					
Annual average power loss (kW)	118.2953	46.0170	35.6760	40.0428	35.1939
Annual average VD (p.u)	0.30662	0.0790	0.0688	0.0753	0.0659
Annual average VSI (p.u)	1.3744	1.1757	1.1359	1.1515	1.1315
Annual average min VSI (p.u)	0.7355	0.8566	0.8856	0.8737	0.8884
Annual average min voltage (p.u)	0.9252	0.9614	0.9696	0.9663	0.9704
Annual average max voltage (p.u)	1	1.0037	1.0016	1.0007	1.0022

Notation: Bold values indicate the best obtained solutions.

Table 7
Performance comparison of algorithms.

Algorithm	Best	Average	Worst	SD
GWO	0.47277	0.48293	0.50171	0.008884
BKA	0.41347	0.425872	0.44654	0.012735
FLO	0.44013	0.45965	0.47262	0.010950
IFLO	0.40834	0.40960	0.41207	0.001232

Notation: Bold values indicate the best obtained solutions.

changes in uncertain factors such as wind speed, solar irradiance, and temperature. Figs. 15 and 16 display the system voltage profiles of both with and without integration of RES. It is obvious from these figures that the voltage profile significantly improved with the incorporation of WT units and solar PV strings. Furthermore, as depicted in Fig. 17, the integration of RES has notably reduced power losses. This reduction is evident across seasons of the year (96-hour period), especially during peak hours, highlighting the positive impact of RES on improving system efficiency and minimizing energy losses. Additionally, Fig. 18 illustrates the effect of RES integration on system voltage stability. The figure clearly shows a marked enhancement in voltage stability throughout seasons of the year, further underscoring the benefits of RES in maintaining stable and reliable voltage levels.

Table 7 presents a performance comparison between the proposed IFLO algorithm and other optimization techniques, including GWO, BKA, and the original FLO. The results clearly show that the proposed approach surpasses other methods in terms of mean, best, and worst values, emphasizing its robustness and superior performance in solving the OPP for RESs. Additionally, Fig. 19 illustrates the convergence curves for the OPP of RESs into the RDG, where IFLO demonstrates a faster convergence rate and achieves a lower fitness value compared to the other techniques. This superior performance is due to the improved strategies incorporated into the proposed algorithm.

8. Conclusion

This research addressed the complex challenge of the optimal planning problem of RESs, specifically WT and PV farms within the distribution grids. Our primary goal was to enhance system performance by minimizing power losses and voltage deviations while improving the voltage stability index, across the entire system. We proposed a novel improved FLO algorithm (IFLO) specifically designed to solve the presented problem. The IFLO incorporates three distinct strategies, including FDB, QOBL and CM to enhance its searching capabilities and avoid it getting trapped in local optima. We tested the IFLO on 23 different functions and compared it to other effective optimization algorithms, such as GWO, HHO, CDO, DOA, WOA, BKA, and the original FLO. The results show that IFLO achieves superior performance compared to conventional algorithms. The optimal locations and rating capacities of PV strings and WTs, along with the optimal PF of WTs, were determined simultaneously. The planning process accounted for seasonal uncertainties in wind speed, solar irradiance, and temperature throughout the year, alongside the seasonal fluctuation in load demand, all within a time-dependent generation-load model. The proposed approach effectively solved the OPP of RESs into the 85-bus distribution grid, reducing power losses, elevating the voltage profile, and improving grid voltage stability. Specifically, IFLO reduces annual average power losses by 70.25%, decreases voltage deviation by 78.51%, and enhances the voltage stability index by 20.79% compared to the base case, outperforming other recent optimization techniques such as GWO, BKA and the original FLO. These results highlight the effectiveness and robustness of the proposed method in addressing the complex challenges of RES planning within the distribution grid, making it a valuable and efficient tool for future applications in modern grid planning. Future research could explore more complex integrations, such as electric vehicle (EV) charging stations alongside RESs in larger distribution grids. We aim to expand the current framework by incorporating additional emerging factors such as cost-effectiveness, resilience under extreme scenarios, and grid capacity expansion. Besides, we plan

to explore and investigate incorporating advanced probabilistic models, such as fuzzy cloud theory and interval analysis, to handle uncertainty effectively.

CRedit authorship contribution statement

Badreddine Bendriss: Writing – original draft, Software, Data curation. **Samir Sayah:** Writing – original draft, Validation, Supervision, Resources, Methodology. **Abdellatif Hamouda:** Writing – review & editing, Validation, Supervision, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix

See Tables A.1–A.3.

Table A.1
Unimodal functions.

Function	Dim	Range	Global $F_{\min}$
$f_1(z) = \sum_{j=1}^n z_j^2$	30	$[-100, 100]$	0
$f_2(z) = \sum_{j=1}^n z_j + \prod_{j=1}^n z_j $	30	$[-10, 10]$	0
$f_3(z) = \sum_{j=1}^n (\sum_{i=1}^j z_i)^2$	30	$[-100, 100]$	0
$f_4(z) = \max_j z_j , \quad 1 \leq j \leq n$	30	$[-100, 100]$	0
$f_5(z) = \sum_{j=1}^{n-1} [100(z_{j+1} - z_j^2)^2 + (z_j - 1)^2]$	30	$[-30, 30]$	0
$f_6(z) = \sum_{j=1}^n (z_j + 0.5)^2$	30	$[-100, 100]$	0
$f_7(z) = \sum_{j=1}^n j z_j^4 + \text{rand}(0, 1)$	30	$[-1.28, 1.28]$	0

Table A.2
Multimodal functions.

Function	Dim	Range	Global $F_{\min}$
$f_8(z) = \sum_{j=1}^n -z_j \sin(\sqrt{ z_j })$	30	$[-500, 500]$	-12569.5
$f_9(z) = \sum_{j=1}^n [z_j^2 - 10 \cos(2\pi z_j) + 10]$	30	$[-5.12, 5.12]$	0
$f_{10}(z) = -20 \exp\left(-0.2\sqrt{\frac{1}{n} \sum_{j=1}^n z_j^2}\right) - \exp\left(\frac{1}{n} \sum_{j=1}^n \cos(2\pi z_j)\right) + 20 + e$	30	$[-32, 32]$	0
$f_{11}(z) = \frac{1}{4000} \sum_{j=1}^n z_j^2 - \prod_{j=1}^n \cos\left(\frac{z_j}{\sqrt{j}}\right) + 1$	30	$[-600, 600]$	0
$Lf_{12}(z) = \frac{\pi}{n} \left\{ 10 \sin^2(\pi x_1) + \sum_{j=1}^{n-1} [(x_j - 1)^2 (1 + 10 \sin^2(\pi x_{j+1}))] + (x_n - 1)^2 \right\} + \sum_{j=1}^n u(z_j, 10, 100, 4)$	30	$[-50, 50]$	0
$Lf_{13}(z) = 0.1 \left\{ \sin^2(3\pi z_1) + \sum_{j=1}^{n-1} (z_j - 1)^2 [1 + \sin^2(3\pi z_j + 1)] + (z_n - 1)^2 [1 + \sin^2(2\pi z_n)] \right\} + \sum_{j=1}^n u(z_j, 5, 100, 4)$	30	$[-50, 50]$	0
$u(z_j, a, b, c) = \begin{cases} b(z_j - a)^c, & z_j > a \\ 0, & -a \leq z_j \leq a \\ b(-z_j - a)^c, & z_j < -a \end{cases}$			

Table A.3
Fixed-dimension multimodal benchmark functions.

Function	Dim	Range	Global $F_{\min}$
$f_{14}(z) = \left[\frac{1}{500} + \sum_{i=1}^{25} \frac{1}{i + \sum_{j=1}^{25} (z_j - a_{ij})^6} \right]^{-1}$	2	$[-65.536, 65.536]$	0.998
$f_{15}(z) = \sum_{j=1}^{11} \left[b_j - \frac{z_j(b_j^2 + b_j z_j)}{b_j^2 + b_j z_j + z_1} \right]^2$	4	$[-5, 5]$	0.00030
$f_{16}(z) = 4z_1^2 - 2.1z_1^4 + \frac{1}{3}z_1^6 + z_1 z_2 - 4z_2^2 + 4z_2^4$	2	$[-5, 5]$	-1.0316
$f_{17}(z) = \left(z_2 - \frac{5.1}{4\pi^2} z_1^2 + \frac{5}{\pi} z_1 - 6 \right)^2 + 10 \left(1 - \frac{1}{8\pi} \right) \cos z_1 + 10$	2	$[-5, 5]$	0.3979
$f_{18}(z) = [1 + (z_1 + z_2 + 1)^2 (19 + 14z_1 + 3z_1^2 - 14z_2 + 6z_1 z_2 + 3z_2^2)] \times [30 + (2z_1 - 3z_2)^2 (18 - 32z_1 + 12z_1^2 - 48z_2 - 36z_1 z_2 + 27z_2^2)]$	2	$[-2, 2]$	3
$f_{19}(z) = -\sum_{j=1}^4 c_j \exp \left[-\sum_{i=1}^3 a_{ji} (z_i - p_{ji})^2 \right]$	3	$[1, 3]$	-3.86
$f_{20}(z) = -\sum_{j=1}^4 c_j \exp \left[-\sum_{i=1}^6 a_{ji} (z_i - p_{ji})^2 \right]$	6	$[0, 1]$	-3.32
$f_{21}(z) = -\sum_{j=1}^5 [(z - a_j)(z - a_j)^T + c_j]^{-1}$	4	$[0, 10]$	-10.1532
$f_{22}(z) = -\sum_{j=1}^7 [(z - a_j)(z - a_j)^T + c_j]^{-1}$	4	$[0, 10]$	-10.4028
$f_{23}(z) = -\sum_{j=1}^{10} [(z - a_j)(z - a_j)^T + c_j]^{-1}$	4	$[0, 10]$	-10.5363

Data availability

Data will be made available on request.

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